

Nevò MIRZAI HAMADANI

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EDUCATION

EPFL | Master in Robotics (2nd year) Sep 2024 – July 2026

- Minor in Data Science: RL: 6/6, DL: 5.75/6, Image analysis: 5.75/6
- Relevant courses: Robotic Manipulation: 5.5/6, Aerial Robotics: 5.5/6

EPFL | Bachelor in Mechanical Engineering Sep 2021 – July 2024

- 1 year exchange at the Chinese University of Hong Kong (CUHK)
- Relevant courses: Linear Algebra: 5.75/6, Analysis: 5.25/6, Algorithms: 5.75/6

TECHNICAL SKILLS

 **Programming** Python, C++, C, MATLAB, C#

 **Simulation** IsaacSim, IsaacLab, OpenAI gym, Unity, Genesis, Webots, MuJoCo, Gazebo

 **Computer Vision** SLAM (Visual/LiDAR), Image Analysis, OpenCV

 **Machine Learning** Reinforcement Learning, Deep Learning, Pytorch

 **Technical** ROS2, CUDA, Git, IK, Optimization, Control

 **Laboratory** Soldering, electrical circuits, CAD, 3D Printing

EXPERIENCE

Student Mentor Sep 2022 – January 2026

- Supporting freshmen during their academic year with advice and practical help.

Teaching Assistant Sep. 2022 – June. 2024

- Mechatronics, Electronics, Linear Algebra, Calculus, Physics I, Thermodynamics, Algorithms, Programming, Mechanical Behaviour of Materials, Numerical Analysis.
- Supporting students during exercise sessions.

EPFL Rocket Team Sep 2021 – June 2024

Vertical landing rocket research project (Icarus) - Simulink

- Simulated vertical landing of our rocket in Simulink. Currently testing for Sim2Real.

RESEARCH PROJECTS

3D Printing Drone Simulation, Sensor Simulation - Waiting for Paper Publication

- Constructed a simulation framework for a 3D printing drone using the Genesis engine.
- Modeled viscoelastic curing and expanding fluids using the Material Point Method by modifying the physics simulation engine at its core.

Hand Gesture Drone Swarms Control - Sim2Real, ROS

- Conceived an intuitive system for controlling a swarm of drones using hand gestures.
- Created and controlled a simulated swarm in Unity, and applied ROS to control a real swarm (Digital-Twin).

Soft Robotic Arm: Sim2Real Digital Twin & RL - Isaac Sim

- Deployed a Digital Twin of a soft robotic arm using NVIDIA Isaac Sim.
- Approximated soft dynamics with rigid-body physics and calibrated stiffness/damping.
- Modeled an IMU sensor behaviour for accurate simulation data.

PROJECTS

Reinforcement Learning for Quadruped Robot - OpenAI Gym, Isaac Gym

- Trained a SAC Reinforcement Learning policy to achieve quadruped robot locomotion, including high/low velocity gait control and stair climbing.
- Investigated and integrated movement algorithms (Central Pattern Generators (CPG)).

Robotic Arm Simulation for Scaffolding Applications in Hong Kong - Isaac Sim

- Simulated inverse kinematics for a robotic arm to assist in scaffolding construction.

Inverse RL research - Reinforcement Learning

- Conducted research into the limitations and possible solutions of current IRL methods.

Autonomous drone racing - Webots

- Programmed a drone to race a given trajectory autonomously, first in simulation and then deployed on a real Crazyflie (Sim2Real). Path planning, Computer Vision.

Billiard Game Analysis through Computer Vision - Simulink

- Assessed billiard game dynamics using MATLAB, C, and LabVIEW to detect the winner.

Sensor Noise Characterization & Synchronous Detection

- Analyzed sensor noise spectrally, validated measurements, and implemented lock-in detection for ambient light rejection.

Autonomous Vehicle Navigation System Using MPC - Simulink

- Devised a simplified autonomous vehicle navigation system in simulation (MPC).

Eye Tracking Headlamp - Robotics Manipulation

- Prototyped a Raspberry Pi-based headlamp using dual IR/optical cameras for real-time gaze tracking.
- Engineered a closed-loop system to automatically align lighting with the user's eyes.

Data Augmentation Box - Object Recognition - YOLO, System Integration

- Developed an automated physical environment to generate diversified organic datasets. Then fine-tuned object recognition models (YOLO).

AI model training for Hate Speech prediction - GPT2/BERT

- Optimized GPT-2 and BERT to predict the probability that hate speech will be used in a phrase given the context, then compared the results.

Clickbait data analysis: Is clickbait recognizable from data? - Pandas, Pytorch

- Analyzed metadata from YouTube and developed indicators of clickbait videos.

Chocolate recognition - Computer Vision

- Leveraged Computer Vision to recognize and count the number of different chocolates in multiple images with varying backgrounds.

Computer Vision Navigation for a Thymio Robot - Robotic Manipulation

- Implemented vision (OpenCV), Kalman filter, and path planning for a Thymio robot.

Quadcopter Design and Stabilization - Robotic Manipulation

- Programmed PID stabilization from scratch using an MPU and an Arduino.

AWARDS - INTERESTS - LANGUAGES

 Guinness World Record (2017): Most drones built and flown simultaneously (U18)

 Judo (19 years experience, pre-professional level)

 English C1, French C1, Italian C2, German B1, Hebrew B2